
COHE: Auditing Non-Transitivity in Sample Difficulty Proxies for Vision Models

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Abstract

Label-free proxy scores are often used as if they measured downstream discriminative difficulty. Ranking agreement, held-out prediction, and downstream benefit are distinct evidence levels. We present *Cross-Objective Hardness Evaluation* (COHE), a protocol that audits a specified proxy, target, and protocol through dependence, predictive-validity, and operational-transfer gates, mapping standard statistics to bounded claim tiers. DINOv2 isolation captures a first-learning relation ($\rho = 0.4363$, $R^2 = 0.2126$), yet direct hard-only selection underperforms Random at 10/30/50% budgets (−14.87/ − 17.25/ − 11.12 percentage points); class-balanced 10% selection also underperforms (−11.40), with all 20 paired seed differences negative. Across CIFAR-10/100, ImageNet-10, and ImageNet-1K scoring checks, eight primary generative proxy families do not support CE/margin surrogate claims. Operational checks include DDPM full-data blocked ordering and ImageNet-1K class-balanced VAE-hard selection at 30%, the latter yielding −7.765 points versus Random (95% CI [−8.170, −7.360]). These findings concern the tested policies and learners. A task-aligned ImageNet retrieval/triage control passes all gates. The public artifact provides audit and recovered experiment code, seven CIFAR-100 proxy scalar arrays with sample-order specifications, aggregate and seed-level results, provenance, and verified author-trained checkpoints for retained endpoints, without redistributing upstream images or weights.

1 Introduction

A reconstruction error, diffusion score, token likelihood, or representation-isolation score can be meaningful for the objective that produced it. The stronger step is to use that score as a ranking of classification difficulty, a data-selection rule, or curriculum evidence. COHE audits this step: what downstream statement is supported by a specified proxy, target, and declared protocol?

Generative scores mainly describe input regularity, whereas discriminative difficulty depends on the label, classifier, and decision protocol. The $P(x)$ versus $P(y \mid x)$ distinction motivates the audit. Here, *non-transitivity* means that rank alignment, held-out predictive validity, and operational benefit do not automatically imply one another; it is an empirical evidence distinction, not a theorem about a binary relation. Gate 1 asks whether the proxy ranks samples like the target; Gate 2 asks whether it predicts held-out target hardness; Gate 3 asks whether its declared use improves the downstream metric against the stated baseline. We reserve actionability for Gate 3. Figure 1 previews these evidence gates.

The DINOv2 stress test exposes this boundary: isolation predicts first-learning ($\rho = 0.4363$, $R^2 = 0.2126$), yet direct hard-only selection fails at 10/30/50% budgets and under a class-balanced 10% control, each with five paired seeds (Table 5). The first-learning relation remains meaningful; these tests do not establish benefits for other curricula, weighting, pacing, architectures, or tasks.

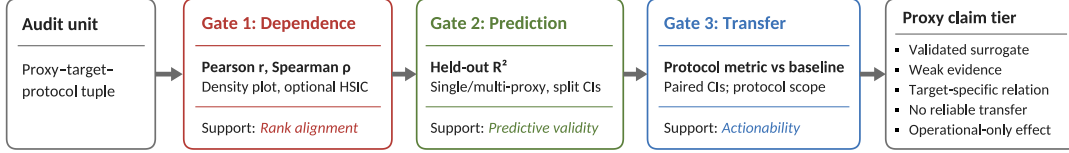


Figure 1: **COHE evidence gates.** COHE audits the tuple through dependence, held-out prediction, and operational-transfer gates. Each gate licenses only the corresponding evidence level; stronger downstream claims require stronger evidence.

The generative-proxy audit is the large-scale calibration case. Across eight primary generative proxy families, CE/margin surrogate claims are unsupported by dependence, prediction, and transfer evidence: correlations are near zero, multi-proxy prediction explains essentially none of CE variance ($R^2 = 0.0002$, 95% CI $[-0.0004, 0.0009]$), and generative-hard selection has no consistent reliable transfer.

COHE contributes three evidence gates with bounded claim wording, a DINOv2 boundary case, a generative-proxy audit including larger-scale and policy-specific operational checks, and scalar arrays, result summaries, audit code, and provenance for verification and reuse.

2 Related work

Discriminative sample difficulty. Many difficulty notions are explicitly tied to a supervised objective. Curriculum learning ranks examples by training difficulty or pedagogical value [Bengio et al., 2009, Soviany et al., 2022]; active learning uses uncertainty or coverage to acquire labels [Settles, 2009, Sener and Savarese, 2018]; pruning uses early training signals or proxy models [Coleman et al., 2020, Paul et al., 2021]; and training-dynamics work measures forgetting, confidence, variability, or ambiguous regions [Toneva et al., 2019, Swayamdipta et al., 2020]. Other task-aligned signals include c-scores [Jiang et al., 2021], prediction depth [Baldock et al., 2021], recurrent visual-processing evidence [Kar et al., 2019], human viewing-time estimates [Mayo et al., 2023], dataset-complexity measures [Rousseeuw, 1987, Cao et al., 2024, 2025], and probe nets [Scheidegger et al., 2021]. These works define or use difficulty relative to a task, label structure, model family, or human behavior. COHE instead audits a cross-objective claim: whether a task-agnostic generative score can support those task-aligned difficulty claims when the objective changes.

Generative and proxy-based evaluation. Generative models provide label-free signals for image structure and likelihood-like regularity. VAEs, diffusion models, latent diffusion models, VQ tokenizers, CLIP-space reconstruction, and autoregressive image models yield reconstruction error, denoising error, latent round-trip discrepancy, semantic mismatch, or token NLL [Kingma and Welling, 2014, Ho et al., 2020, Song et al., 2021, Radford et al., 2021, Rombach et al., 2022, Chen et al., 2020, Wang et al., 2024]. Such signals have been studied for OOD detection, typicality, likelihood regret, filtering, and augmentation [Nalisnick et al., 2019a,b, Xiao et al., 2020, Trabucco et al., 2023]. They may be meaningful for $P(x)$ -style quality, likelihood, reconstruction, or distributional behavior, but that does not by itself validate their use as $P(y | x)$ -style discriminative rankings.

Evaluation-audit framing. COHE evaluates the claim made from a score. As an evaluation and reporting protocol, it specifies what evidence should be reported before a proxy-induced ranking is described as a discriminative difficulty surrogate. Its contribution is the separation of evidence levels and the conversion of heterogeneous results into bounded claim tiers. COHE differs from work that proposes a new score, evaluates a selector only by downstream accuracy, or reports correlations without mapping evidence to allowed claims. A downstream gain can support an operational policy claim without proving scalar-surrogate validity, and a correlation can support alignment without proving actionability. This separation of evidence and claims is the resource contributed here.

COHE is also related to surrogate endpoint validation in biostatistics, where association with an intermediate endpoint is insufficient to justify a downstream endpoint claim [Prentice, 1989, Freedman et al., 1992]. Whereas those criteria concern population-level causal mediation, COHE audits sample-level ML settings using dependence, held-out prediction, and operational transfer.

Taken together, prior work supplies many useful difficulty and proxy scores; the missing step is auditing when evidence for one objective supports a cross-objective downstream claim. COHE is motivated by that reporting mismatch. A label-free score can be valid for reconstruction quality, likelihood-like regularity, OOD behavior, filtering heuristics, augmentation, or representation isolation, while still requiring separate evidence before it supports a CE/margin surrogate, curriculum, filtering-benefit, or actionability claim. Appendix Table 11 summarizes how COHE complements prior surrogate-endpoint and proxy-selection frameworks. Section 3 formalizes the scalar-surrogate assumption behind such escalations.

3 Problem setup: The scalar-surrogate assumption

Let $x \in \mathcal{X}$ be an image and $y \in \mathcal{Y}$ its label. A task-agnostic proxy assigns a scalar $D_{\text{gen}}(x)$, such as denoising or reconstruction error, latent or VQ discrepancy, token NLL, or DINOv2 isolation. A discriminative target assigns $D_{\text{disc}}(x, y; f)$ for classifier f , such as CE, margin, GradNorm, forgetting, or first-learning. For diagnostic visualizations, $(D_{\text{gen}}, D_{\text{disc}})$ is an audit space rather than evidence for exact geometric independence. A scalar proxy used as a surrogate implicitly assumes that some monotone function g makes $D_{\text{disc}}(x, y; f) \approx g(D_{\text{gen}}(x))$ for most samples, at least well enough to support ranking, prediction, or downstream selection.

Scalar-surrogate validity. Using a task-agnostic proxy as a surrogate for D_{disc} requires cross-objective validity for the downstream target. We distinguish three increasingly operational notions:

1. **Rank alignment:** D_{gen} shows a declared-threshold monotone association with D_{disc} .
2. **Predictive validity:** D_{gen} , or a vector of generative proxies, predicts D_{disc} on held-out samples.
3. **Operational transfer:** using D_{gen} yields downstream benefit relative to appropriate baselines under a specified protocol.

When this scalar-surrogate assumption fails, COHE limits the downstream claim rather than discarding the score: the proxy may still capture a target-specific channel, such as GradNorm sensitivity or first-learning dynamics, without becoming a general discriminative-hardness score.

4 COHE: Cross-Objective Hardness Evaluation

COHE does not return a single pass/fail score. Instead, it reports the strongest bounded claim supported by the evidence in a specified proxy–target–protocol setting. Each *audited setting* specifies the score, target definition, downstream use, dataset, learner, splits, and baselines. Gate 1 supports alignment, Gate 2 adds held-out predictive validity, and Gate 3 supports protocol-specific actionability (Table 1). Table 2 maps the resulting evidence patterns to six reporting tiers; Appendix Table 13 gives their full definitions and precedence.

Thresholds depend on target scale, sample size, and intervention cost. The descriptive $R^2 \leq 0.005$ near-zero band is local to this audit, not a universal constant. CE/margin and final DINOv2 claim conclusions are unchanged over candidate bands $\{0.001, 0.0025, 0.005, 0.010\}$ (Appendix Table 10).

Threshold calibration. For a new audit, record the proxy, target, protocol, score direction, split seeds, baselines, candidate thresholds, and safety margin before tier assignment. Calibrate against matched shuffled-proxy behavior and split or paired-baseline variability, then freeze the specification. These are prospective requirements: retained historical records do not establish a predeclared numerical G1 threshold or G2 safety margin for every setting. The near-zero band and sensitivity check are descriptive. Appendix A.1 separates this procedure from the retained evidence.

G3 evaluates the declared downstream endpoint against its matched baseline. For subset retraining, we require a seed-paired accuracy-difference CI entirely above zero and benefit stable over the declared scope; retrieval/triage instead compares target-hard recall with a shuffled-ranking null (Appendix Table 36). A lower-cost audit may use a fixed learner, policy/budget grid, and paired seeds, reported as *pilot operational evidence*. If G3 is infeasible, stop at the G1/G2-supported claim. An unmeasured gate is not a failure, and a failed policy does not erase auxiliary-target relations.

Table 3 lists representative claim-risk patterns; COHE specifies which claim is supported by the reported evidence.

Table 1: COHE protocol.

Gate	Question	Minimum evidence	Supports
G1: Dependence	Does the proxy align with the target?	Pearson r, Spearman ρ ; density plot; optional HSIC	Rank alignment
G2: Prediction	Can it predict held-out target hardness?	Held-out R^2 ; single/multi-proxy; split CIs	Predictive validity
G3: Transfer	Does using it help?	Protocol metric vs baseline ; endpoint uncertainty; declared scope	Protocol-scoped actionability

Table 2: COHE gate rules and tier mapping.

<i>Evidence gates</i>			<i>Tier mapping</i>	
Gate	Rule	Licenses	Evidence pattern	Tier
G1	Association $>$ null + threshold	Alignment	G1+G2+G3	Validated surrogate
G2	Held-out R^2 CI $>$ baseline + margin	Predictive validity	G1+G2; no G3 pass	Predictive validity only
G3	Endpoint-specific gain vs baseline; protocol-stable	Actionability	Auxiliary relational signal; no G3 pass	Target-specific relation
			G3; G1/G2 not both positive	Operational-only effect
			Incomplete or inconclusive	Weak evidence
			No positive G1/G2; weak G2 / failed G3	No reliable transfer

Table 3: Claim-risk patterns.

Pattern	Risky escalation	COHE-bounded claim
Reconstruction / denoising	$P(x)$ regularity $\rightarrow$ CE/margin difficulty	Alignment or $P(x)$ -regularity only unless G2 passes
Generative filtering	Transfer result $\rightarrow$ scalar-surrogate validity	Operational-only effect unless G1–G2 pass
Token-NLL quality	Likelihood $\rightarrow$ task difficulty	$P(x)$ -quality only without target audit
Representation isolation	Dynamics prediction $\rightarrow$ curriculum/filtering claim	Target-specific relation unless matched G3 passes
Task-aligned ranking	Rank agreement $\rightarrow$ actionability	Gate-1 anchor; G2/G3 needed for stronger claims

What COHE is useful for. COHE applies when a convenient label-free score is about to support a downstream discriminative claim: filtering by reconstruction error, ranking by denoising difficulty, using tokenizer or autoregressive likelihood as data quality, or treating visually hard samples as classification-informative. It does not ask whether the proxy is meaningful for $P(x)$; it asks what discriminative claim the proxy supports and returns a bounded claim tier for the stated target and protocol.

5 Experimental setup

Datasets and targets. We audit CIFAR-10/100 [Krizhevsky and Hinton, 2009], ImageNet-10 (Imagenette) [Howard, 2019], and ImageNet-1K [Deng et al., 2009]. CIFAR scoring uses the full 50k training sets; operational checks train ResNet-18 [He et al., 2016] unless stated otherwise. ImageNet-1K scoring uses all 50,000 validation images with pretrained ResNet-50 targets. Its separate Gate 3 trains ResNet-18 from scratch for 90 epochs on class-balanced 30% train subsets, using five paired seeds and the same best-validation-Top-1 checkpoint rule in both arms. ImageNet-10 uses train-split VAE/CE selection with ResNet-18 head fine-tuning. Targets include CE, margin, GradNorm, forgetting, and first-learning. Table 4 maps the main blocks; Appendix Table 7 gives the complete roadmap.

Proxy families. The eight primary generative proxy families are DDPM score norm, DDIM reconstruction, DDIM-CLIP, VAE reconstruction, SD-VAE latent round-trip, SD-v1.5 latent denoise, Emu3 VQ reconstruction, and ImageGPT token NLL. We use “generative hardness” as an operational umbrella for label-free reconstruction, denoising, tokenizer, latent, semantic, or likelihood-like scores, not as a claim that all estimate one quantity; formal definitions and auxiliary checks are in Appendix

Tables 15–16. The CE/margin conclusion does not rely on one checkpoint family: the same low-surrogate-validity pattern appears for latent, VQ, semantic, autoregressive, ImageNet-1K validation, and CIFAR-100-trained ConvAE checks (Appendix Table 18).

Representation-level stress test. To test whether the low CE/margin pattern is specific to generation-objective proxies, we include DINOv2 isolation as a non-generative boundary stress test rather than a representative generative proxy [Oquab et al., 2024]. The proxy is the mean cosine distance to the $k = 20$ nearest DINOv2 CLS neighbors; labels are not used. Its role is to test whether COHE preserves a target-specific Gate-2 relation while preventing escalation to an unmatched Gate-3 actionability claim.

Table 4: **Experimental map of the COHE audit.** Related settings are grouped; Appendix Table 7 gives the complete roadmap.

Dataset / split	Proxy or policy	Target / endpoint	Gates	Learner / protocol	Main finding
CIFAR-100 train	DDPM, DDIM(-CLIP), SD-VAE, SD-v1.5, Emu3; ConvAE check	CE, margin, GradNorm	1–2	ResNet-18 targets; held-out prediction	CE/margin near-zero; GradNorm target-specific
ImageNet-1K val	VAE, SD-VAE, SD-v1.5, Emu3 VQ, ImageGPT	CE, margin, VQ, GradNorm	1–2	Pretrained ResNet-50; 50k images	Weak CE/margin; larger GradNorm relations
CIFAR-100 train/test	DINOv2 isolation; Hard	First-learning / Top-1	1–3	ResNet-18; five pairs; 10/30/50%; balanced 10%	G1/G2 signal; all 20 G3 deltas negative
CIFAR-100 train/test	DDPM Gen-Hard; selection controls	Top-1	3	Five pairs; ResNet-18; 10/30/50/70%; ResNet-50: 10/50/70%	No stable Random-beating Gen-Hard transfer
CIFAR-100 train/test	DDPM blocked Easy-to-Hard; shuffled order	Top-1	3	ResNet-18; full data; 10 bins; eight pairs	No positive transfer; true and shuffled order not distinguished
ImageNet-10 train/val	VAE reconstruction; Hard	Top-1	3	ResNet-18 head fine-tuning; three pairs; 10/30%	Protocol-dependent evidence
ImageNet-1K train/val	VAE reconstruction; balanced Hard	Top-1	3	ResNet-18 from scratch; 90 epochs; five pairs; 30%	−7.765 pp vs balanced Random
ImageNet-1K val; CIFAR-100 train	ResNet-18 CE; shuffled DDIM	ResNet-50 CE / triage; CE/margin	1–3; 1–2	Pretrained targets; retrieval control; five shuffled seeds	Task-aligned control passes; shuffled proxy near-null

6 Results

The results keep three evidence levels separate: sample-wise alignment, held-out prediction, and downstream action under a specified protocol. The organizing question is whether evidence at one level can be escalated into a stronger downstream claim.

6.1 Boundary case: Prediction does not imply actionability

DINOv2 isolation has a meaningful first-learning signal ($\rho = 0.4363$, $R^2 = 0.2126$). Its CE association is negative ($\rho = -0.2006$, $R^2 = 0.0108$), and margin prediction is small ($R^2 = 0.0059$); these exceed the descriptive 0.005 band but are much weaker than first-learning. Direct hard-only CIFAR-100 selection with ResNet-18 underperforms Random at 10/30/50% budgets; class-balanced hard selection also underperforms class-balanced Random at 10% (Table 5). All 20 differences across four conditions with five paired seeds each are negative; these are not 20 independent repetitions. The

first-learning tier remains Target-specific relation. The selection tests do not establish benefits for weighting, adaptive pacing, other DINO interventions, architectures, or tasks.

Table 5: **DINOv2 boundary Gate 3 audit.**

Selection / baseline	Budget	Mean Δ Top-1 (pp), 95% CI
Hard / Random	10%	-14.87 [-15.62, -14.12]
Hard / Random	30%	-17.25 [-17.99, -16.52]
Hard / Random	50%	-11.12 [-11.54, -10.70]
Balanced Hard / balanced Random	10%	-11.40 [-12.91, -9.89]

Note. Four conditions with five paired seeds each; paired Student- t CIs. All 20 differences are negative. First-learning evidence supports Target-specific relation. Seed-level results and the separate three-seed reverse-isolation check are in Appendix F.

6.2 Generative proxies weakly support CE/margin surrogate claims

Figure 2 shows representative densities and the proxy-family summary. Panel (a)’s DDIM-CE values are in Appendix Table 20; panels (b–c) are in Table 17. CE and margin remain weakly aligned across CIFAR-100 and ImageNet-1K validation. Larger negative GradNorm associations, especially for VAE, SD-v1.5, Emu3 VQ, and ImageGPT at scale, remain target-specific and do not extend to CE or margin.

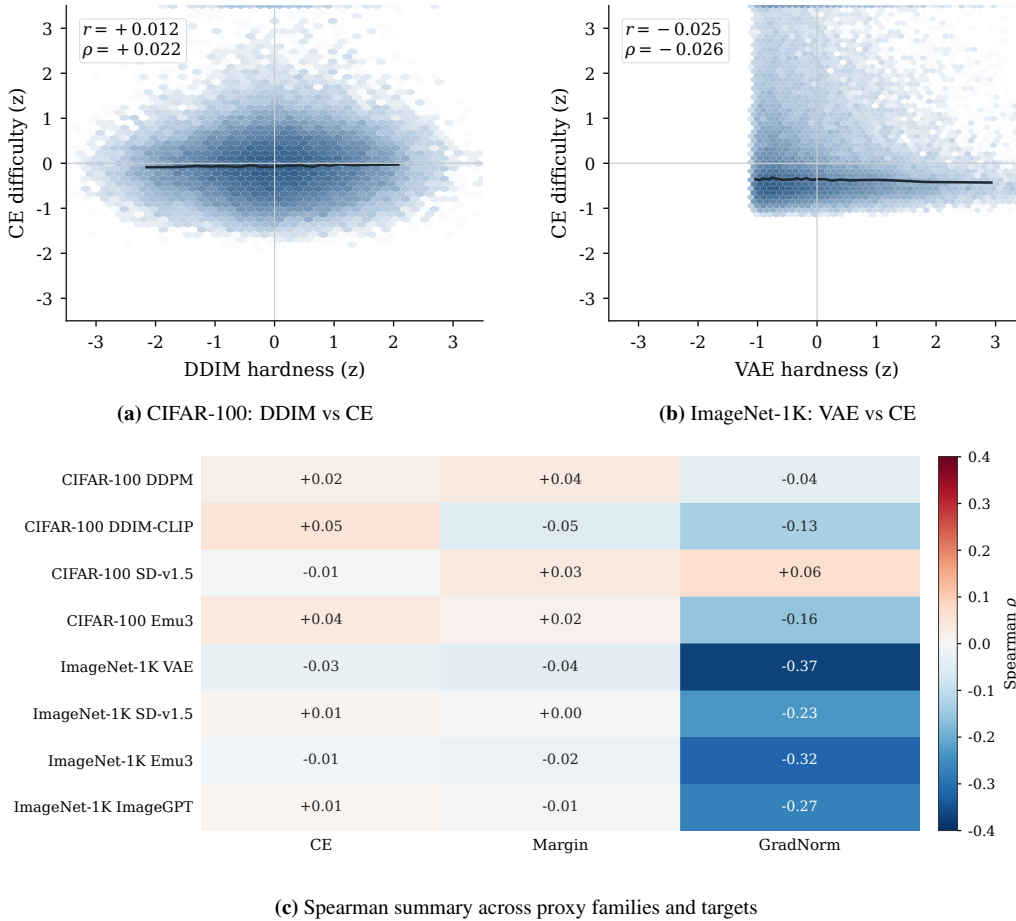


Figure 2: **Dependence audit.** Panels (a–b) show representative generative-hardness vs. CE densities with nearly flat running medians; axes are z-scored and clipped to ± 3.5 for visualization only. Panel (c) summarizes Spearman associations: CE and margin remain weakly aligned, while larger GradNorm associations are target-specific.

An auxiliary RBF-kernel HSIC test [Gretton et al., 2005] pairs the DDPM score norm with CE from the pretrained ResNet-18 source classifier on a seed-0, 5,000-sample subset of CIFAR-100 train. With 200 permutations it is non-significant ($p = 0.612$; Appendix Table 22), consistent with weak measured dependence but not exact independence. Cross-architecture target rank correlations remain 0.856–0.914 (Table 35).

GradNorm associations should be interpreted as target-specific sensitivity-channel evidence. Reconstruction, likelihood-like, or representation-space scores may co-vary with input-sensitivity structure without preserving CE or margin orderings. We do not route these GradNorm associations through the accuracy-selection Gate 3 used for CE/margin, because the endpoint would not match the sensitivity target. Under COHE, a GradNorm-oriented Gate 3 would need a sensitivity-specific downstream endpoint, such as robustness, pruning, or gradient-sensitivity reduction. We therefore report the observed associations as target-specific sensitivity evidence, not as CE/margin surrogate validity or subset-selection actionability.

6.3 Held-out prediction preserves the same CE/margin boundary

Gate 2 asks whether a proxy, or proxy vector, predicts discriminative hardness on held-out samples. Figure 3 summarizes the CIFAR-100 split audit; exact values are in Appendix Table 23.

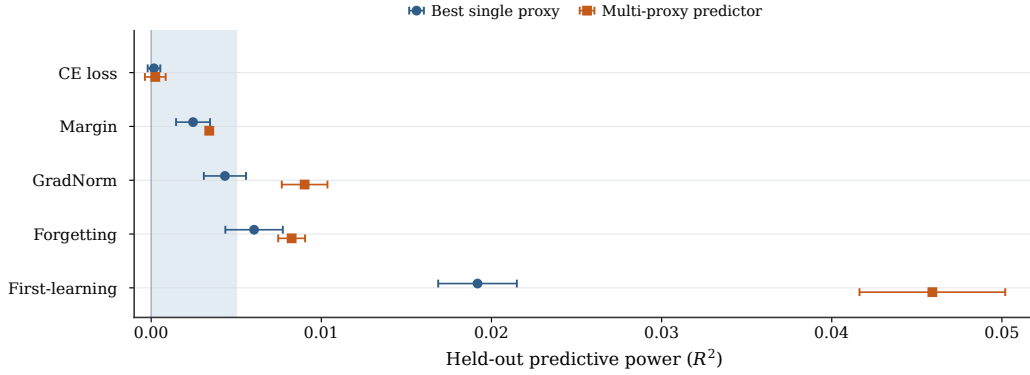


Figure 3: **Predictive-validity audit.** Held-out R^2 remains near zero for CE and margin in the generative-proxy audit even with nonlinear multi-proxy prediction. The shaded region marks the audit-specific near-zero band, $R^2 \leq 0.005$. Larger first-learning values indicate target-specific signal rather than a unified surrogate.

For the audited generative proxy set, CE remains effectively non-predictive even with multi-proxy nonlinear models. Margin improves only to $R^2 = 0.0034$. First-learning is largest, but $R^2 = 0.0459$ still explains less than 5% of held-out variance and is target-specific.

In a separate CIFAR-100/ResNet-18 dynamics audit, the single DDPM score-norm proxy had $r = -0.0053$, $\rho = -0.0064$, and best held-out $R^2 = 0.0000$ for forgetting; for first-learning it had $r = -0.0056$, $\rho = -0.0048$, and best $R^2 = 0.0000$. The epoch-wise audit instead uses DDIM main: maximum absolute Spearman values for cumulative forgetting were 0.0704 under DDIM and 0.0784 under SD-v1.5 (Appendix Tables 39–40). Different proxy sets and split protocols distinguish these checks from the multi-proxy first-learning fit. These audits concern scalar or low-dimensional inputs, not all information in the full representations.

6.4 Operational transfer is protocol-scoped

Gen-Hard ranks CIFAR-100 training samples by the DDPM score-norm proxy: the google/ddpm-cifar10-32 UNet predicts noise after noising images at fixed $t = 500$, and the score is the predicted-noise tensor’s L2 norm, with higher scores treated as harder (Appendix D). ResNet-18 selection uses budgets 0.1/0.3/0.5/0.7 and five paired seeds. Disc-Hard ranks by CE, Predictive Entropy by uncertainty, Coreset-kCenter by coverage, and OHS-XOR samples cases hard on exactly one axis. Figure 4 and Appendix Table 28 show no protocol-stable Random-beating Gen-Hard transfer; the other policies provide coverage and hard-only diagnostics.

This diagnostic is not explained by raw class imbalance alone: class-balanced controls restore class coverage, but balanced Disc-Hard remains below balanced Random, while balanced Gen-Hard has intervals crossing zero and does not yield protocol-stable transfer (Appendix Tables 33 and 34).

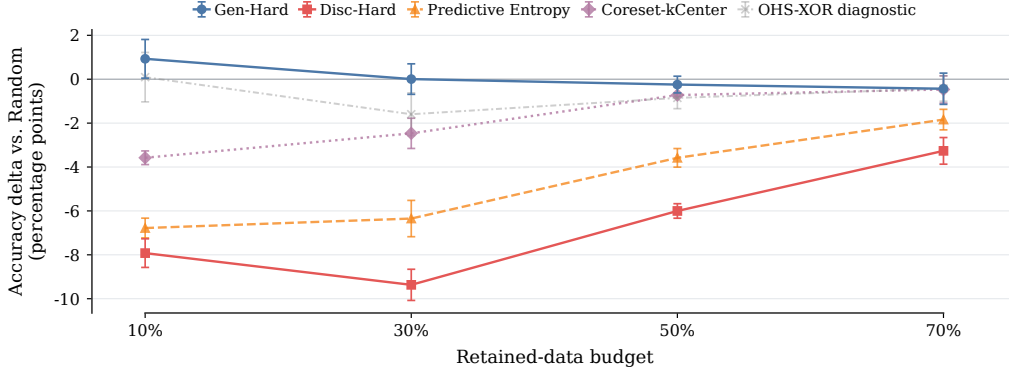


Figure 4: **Operational-transfer audit.** CIFAR-100/ResNet-18 paired deltas versus Random, five seeds at budgets 0.1/0.3/0.5/0.7. Gen-Hard uses the DDPM predicted-noise L2 norm (google/ddpm-cifar10-32, fixed $t = 500$; higher is harder; Appendix D). It has no protocol-stable positive transfer; Disc-Hard and OHS-XOR are diagnostics.

A second-learner CIFAR-100 ResNet-50 audit and a limited ImageNet-10/Imagenette operational check under a fine-tuning protocol give the same qualitative reading: Gen-Hard does not provide consistent reliable transfer across budgets, Disc-Hard remains harmful or protocol-dependent, and negative Gen-Hard deltas in the high-accuracy, low-class-count ImageNet-10 setting are protocol-dependence evidence rather than a stronger proxy-level negative claim (Appendix Tables 29 and 30). In such a fine-tuning protocol, selected hard samples may be atypical rather than broadly informative, so COHE reports protocol-scoped actionability rather than a proxy-level verdict.

To check that the conclusion is not merely an artifact of hard-only rankings, we also evaluated scalar mixtures of generative and discriminative hardness. These mixtures change the operating point but do not yield a stable Random-beating policy, so we treat them as auxiliary diagnostics rather than selection contributions (Appendix Table 32).

Two further checks delimit operational scope. Full-data DDPM Easy-to-Hard ordering uses a fixed 10-bin schedule, all 50,000 samples once per epoch, matched step counts, and eight paired seeds. It yields -0.77 pp versus Random, but is not statistically distinguished from Shuffled-Proxy Order (-0.27 pp, 95% CI $[-0.82, 0.28]$), so harm cannot be uniquely attributed to true score-sample correspondence (Appendix F.1). ImageNet-1K class-balanced VAE-hard selection at 30% yields -7.765 pp versus balanced Random (95% CI $[-8.170, -7.360]$), negative in all five pairs, for ResNet-18 trained from scratch for 90 epochs (Appendix F.2). Neither check licenses conclusions for other policies, proxies, budgets, or learners.

6.5 Hard-only diagnostic: coverage without constraints

The hard-only diagnostic is not a contradiction of discriminative difficulty: highest-loss examples need not form a representative or learnable subset. At budget 0.1, Disc-Hard keeps 94/100 classes, lowers class entropy, puts the subset in the top-10% discriminative tail, and increases joint high-hardness concentration; the pattern persists at budget 0.3 (Appendix Table 31). The class-balanced diagnostics preserve this interpretation while reducing coverage confounding (Appendix Tables 33 and 34). COHE treats these as operational diagnostics, not scalar CE-surrogate evidence.

6.6 Controls and scope

Controls calibrate target stability, matched positive evidence, and null behavior. Task-aligned discriminative hardness rankings retain strong cross-architecture structure on CIFAR-100 (Spearman 0.856 – 0.914 ; Appendix Table 35), whereas the audited task-agnostic proxies remain weakly aligned with those targets. We also include a positive calibration setting on ImageNet-1K validation: ResNet-18 CE used as a proxy for ResNet-50 CE has $r = 0.752$, $\rho = 0.578$, held-out $R^2 = 0.5637$ (95% CI $[0.5578, 0.5695]$; shuffled-null 97.5th percentile < 0.0001), and retrieves 62.48% of the target-hard

top 10% samples, a $6.25\times$ random enrichment (Appendix Table 36). This validates only the declared task-aligned retrieval/triage claim; it is a different setting from the CIFAR-100 subset-retraining audit and does not imply generative-proxy validity or subset-retraining actionability. A shuffled-proxy negative control permutes CIFAR-100 DDIM scores over five seeds; CE, margin, and GradNorm stay near zero (Appendix Table 37), matching the intended null baseline.

CIFAR-100 permits multi-budget, second-learner, ordering, and diversity checks. ImageNet-1K contributes scoring-scale evidence, task-aligned retrieval/triage calibration, and the completed VAE-hard 30% ResNet-18 retraining check. Because the CIFAR-100 discriminative targets are defined on training samples, this audit does not rule out target-side memorization effects; the ImageNet-1K validation analysis provides complementary non-training-split evidence. Each operational claim remains specific to its reported selection or ordering protocol; the limited ImageNet-10 fine-tuning check has a separate scope.

7 Discussion and conclusion

Table 6 summarizes the supported claims.

Table 6: **COHE audit card.**

Use / target	Evidence	Tier	Use boundary
Gen-Hard filtering	G1 weak; G2 near-zero; G3 fails	No reliable transfer	requires task-specific validation
GradNorm	sensitivity association	Target-specific relation	robustness/pruning endpoint
DINOv2 isolation	first-learning signal; all 20 hard-selection deltas negative	Target-specific relation	requires matched Gate 3
Task-aligned rank	cross-arch. $\rho = .856-.914$	Weak evidence (G1 only)	G2/G3 audit
Hard-only selection	low-budget harm; concentration	No reliable transfer	coverage diagnostic

Example claim card. For DINOv2 isolation, G1/G2 support first-learning ($\rho = 0.4363$, $R^2 = 0.2126$); ResNet-18 hard-only selection fails at 10/30/50% and in the balanced 10% control. The claim card therefore records Target-specific relation, with no positive Gate 3 transfer for these policies. It does not assess sample weighting, progressive or adaptive pacing, or other curricula.

COHE’s main lesson is that evidence levels are not interchangeable. Alignment, held-out prediction, and downstream utility support different claims: a positive transfer result alone supports an Operational-only effect, and a failed transfer gate limits the downstream decision claim without erasing target-specific relations. A target-specific tier is therefore a protocol-design pointer, identifying which matched Gate 3 audit would be appropriate next rather than granting that claim in advance.

DDIM-CLIP, timestep sweeps, the 20-superclass check, distinct qualitative hardness regions, and larger target-specific GradNorm associations are consistent with scores emphasizing reconstruction structure, semantic mismatch, texture/background complexity, typicality, or representation geometry that need not preserve a classifier’s decision-boundary ordering. This is a supported hypothesis, not a proven universal mechanism. Hard-only degradation is also consistent with the need for coverage emphasized in coreset and pruning work [Sener and Savarese, 2018, Coleman et al., 2020, Paul et al., 2021]: class balancing restores counts but retains hard-example concentration. Class imbalance alone is insufficient, and the experiments do not isolate noise, outliers, or any single cause. The DDPM ordering control further cautions that blocked scheduling itself can matter; these tests do not assess progressive pacing, adaptive curricula, sample weighting, hard-example mining, or other scheduling policies.

Artifact and reproducibility. The public COHE artifact provides gate-analysis and recovered experiment code, seven CIFAR-100 proxy arrays with sample-index mappings, aggregate and seed-level results, verification utilities, provenance metadata, and 234 verified author-trained checkpoints for retained reported endpoints. Raw datasets, ImageNet-1K score arrays, and third-party pretrained weights are not redistributed and must be obtained from their original sources. Some historical auxiliary experiments have partial code or checkpoint coverage; Appendix J.1 gives the public repository, fixed release version, and exact coverage boundaries.

Limitations, scope, and impact. Scope is standard vision classification, including CIFAR-100, ImageNet-10 fine-tuning, and one ImageNet-1K VAE-hard 30% ResNet-18 retraining check. The proxies are not exhaustive, and these results do not extend to other architectures, interventions, language, video, detection, segmentation, or multimodal pipelines. Gate 3 can require expensive retraining; pilot evidence and G1/G2-only claims must retain their narrower scope. The positive task-aligned retrieval control does not establish generative-proxy validity. Explicit proxy validation may reduce unsupported filtering decisions; unvalidated selection may distort class coverage and downstream datasets, so a difficulty score should not be treated as a fairness guarantee.

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References

- Robert JN Baldock, Hartmut Maennel, and Behnam Neyshabur. Deep learning through the lens of example difficulty. In *Advances in Neural Information Processing Systems*, volume 34, pages 10876–10889, 2021.
- Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*, pages 41–48, 2009.
- Bryan Bo Cao, Abhinav Sharma, Lawrence O’Gorman, Michael Coss, and Shubham Jain. A lightweight measure of classification difficulty from application dataset characteristics. In *International Conference on Pattern Recognition*, pages 439–455, 2024.
- Bryan Bo Cao, Lawrence O’Gorman, Michael Coss, and Shubham Jain. Few-class arena: A benchmark for efficient selection of vision models and dataset difficulty measurement. In *International Conference on Learning Representations*, 2025.
- Mark Chen, Alec Radford, Rewon Child, Jeffrey Wu, Heewoo Jun, David Luan, and Ilya Sutskever. Generative pretraining from pixels. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 1691–1703, 2020.
- Cody Coleman, Christopher Yeh, Stephen Mussmann, Baharan Mirzasoleiman, Peter Bailis, Percy Liang, Jure Leskovec, and Matei Zaharia. Selection via proxy: Efficient data selection for deep learning. In *International Conference on Learning Representations*, 2020.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Fei-Fei Li. Imagenet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.
- Laurence S. Freedman, Barry I. Graubard, and Arthur Schatzkin. Statistical validation of intermediate endpoints for chronic diseases. *Statistics in Medicine*, 11(2):167–178, 1992. doi: 10.1002/sim.4780110204.
- Arthur Gretton, Olivier Bousquet, Alex Smola, and Bernhard Schölkopf. Measuring statistical dependence with hilbert-schmidt norms. In *International Conference on Algorithmic Learning Theory*, pages 63–77. Springer, 2005.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.

- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.
- Jeremy Howard. Imagenette: A smaller subset of 10 easily classified classes from imagenet. <https://github.com/fastai/imagenette>, March 2019. GitHub repository.
- Ziheng Jiang, Chiyuan Zhang, Kunal Talwar, and Michael C. Mozer. Characterizing structural regularities of labeled data in overparameterized models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 5034–5044, 2021.
- Kohitij Kar, Jonas Kubilius, Kailyn Schmidt, Elias B. Issa, and James J. DiCarlo. Evidence that recurrent circuits are critical to the ventral stream’s execution of core object recognition behavior. *Nature Neuroscience*, 22:974–983, 2019. doi: 10.1038/s41593-019-0392-5.
- Diederik P Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations*, 2014.
- Alex Krizhevsky and Geoffrey Hinton. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- David Mayo, Jesse Cummings, Xinyu Lin, Dan Gutfreund, Boris Katz, and Andrei Barbu. How hard are computer vision datasets? calibrating dataset difficulty to viewing time. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, Dilan Gorur, and Balaji Lakshminarayanan. Do deep generative models know what they don’t know? In *International Conference on Learning Representations*, 2019a.
- Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, and Balaji Lakshminarayanan. Detecting out-of-distribution inputs to deep generative models using typicality. *arXiv preprint arXiv:1906.02994*, 2019b. doi: 10.48550/arXiv.1906.02994.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- Mansheej Paul, Surya Ganguli, and Gintare Karolina Dziugaite. Deep learning on a data diet: Finding important examples early in training. In *Advances in Neural Information Processing Systems*, volume 34, pages 20596–20607, 2021.
- Ross L. Prentice. Surrogate endpoints in clinical trials: Definition and operational criteria. *Statistics in Medicine*, 8(4):431–440, 1989. doi: 10.1002/sim.4780080407.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022.
- Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7.
- Florian Scheidegger, Roxana Istrate, Giovanni Mariani, Luca Benini, Costas Bekas, and Cristiano Malossi. Efficient image dataset classification difficulty estimation for predicting deep-learning accuracy. *The Visual Computer*, 37:1593–1610, 2021. doi: 10.1007/s00371-020-01922-5.

- Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations*, 2018.
- Burr Settles. Active learning literature survey. Computer Sciences Technical Report 1648, University of Wisconsin–Madison, 2009.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.
- Petru Soviany, Radu Tudor Ionescu, Paolo Rota, and Nicu Sebe. Curriculum learning: A survey. *International Journal of Computer Vision*, 130(6):1526–1565, 2022.
- Swabha Swayamdipta, Roy Schwartz, Nicholas Lourie, Yizhong Wang, Hannaneh Hajishirzi, Noah A Smith, and Yejin Choi. Dataset cartography: Mapping and diagnosing datasets with training dynamics. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 9275–9293, 2020.
- Mariya Toneva, Alessandro Sordoni, Remi Tachet des Combes, Adam Trischler, Yoshua Bengio, and Geoffrey J Gordon. An empirical study of example forgetting during deep neural network learning. In *International Conference on Learning Representations*, 2019.
- Brandon Trabucco, Kyle Doherty, Max Gurinas, and Ruslan Salakhutdinov. Effective data augmentation with diffusion models. In *ICLR Workshop on Mathematical and Empirical Understanding of Foundation Models*, 2023.
- Xinlong Wang, Xiaosong Zhang, Zhengxiong Luo, Quan Sun, Yufeng Cui, Jinsheng Wang, Fan Zhang, Yuezhe Wang, Zhen Li, Qiying Yu, Yingli Zhao, Yulong Ao, Xuebin Min, Tao Li, Boya Wu, Bo Zhao, Bowen Zhang, Liangdong Wang, Guang Liu, Zheqi He, Xi Yang, Jingjing Liu, Yonghua Lin, Tiejun Huang, and Zhongyuan Wang. Emu3: Next-token prediction is all you need. *arXiv preprint arXiv:2409.18869*, 2024.
- Zhisheng Xiao, Qing Yan, and Yali Amit. Likelihood regret: An out-of-distribution detection score for variational auto-encoder. In *Advances in Neural Information Processing Systems*, volume 33, pages 20685–20696, 2020.
- Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. In *Proceedings of the British Machine Vision Conference*, 2016.

Appendix

This appendix provides the reusable COHE audit checklist, formal proxy definitions, additional dependence and predictive-validity results, operational-transfer diagnostics, and reproducibility details.

Experiment roadmap

Table 7: **Complete experiment roadmap: core audits and operational checks.** Selection scores use training samples; test or validation endpoints are evaluated separately.

Location	Dataset / split	Proxy	Target	Gates	Learner	Seeds / budgets	Result / purpose
§6.2; Tables 17, 20	CIFAR-10/100 train; ImageNet-10 val	DDPM, DDIM, SD-v1.5, Emu3	CE, margin, GradNorm	1	ResNet-18	Full train; 3,925 ImageNet-10 val	Dependence audit; weak CE/margin
Table 21	ImageNet-1K val	VAE, SD-VAE, SD-v1.5, Emu3, ImageGPT	CE, margin, GradNorm	1–2	Pretrained ResNet-50	50,000 images	Scoring scale; GradNorm target-specific
§6.3; Table 23	CIFAR-100 train, held-out splits	Single / multi-proxy scores	CE, margin, dynamics	2	ResNet-18; lin-ear/nonlinear fits	Five split seeds	Weak held-out CE/margin prediction
Table 18	CIFAR-100 train	CIFAR-100 ConvAE	CE, margin, GradNorm, dynamics	1–2	ConvAE / ResNet-18	AE seed 0; 50 epochs	Dataset-specific reconstruction control
§6.1; Tables 19, 24	CIFAR-100 train/test	DINOv2 isolation	First-learning / Top-1	1–3	DINOv2-small / ResNet-18	Five pairs; 10/30/50%; balanced 10%	Target-specific signal; 20 negative hard-selection deltas
§6.4; Table 28	CIFAR-100 train/test	DDPM Gen-Hard; CE, entropy, kCenter, OHS-XOR	Top-1	3	ResNet-18	Five pairs; 10/30/50/70%	Main operational selection
Table 29	CIFAR-100 train/test	DDPM, CE, OHS-XOR	Top-1	3	ResNet-50	Five pairs; 10/50/70%	Second-learner check
Table 30	ImageNet-10 train/val	VAE, CE, OHS-XOR	Top-1	3	ResNet-18 head fine-tuning	Three pairs; 10/30%	Limited protocol-dependent check
§F.2	ImageNet-1K train/val	VAE; balanced Hard	Top-1	3	ResNet-18 from scratch	Five pairs; 30%; 90 epochs	Larger-scale selection check
§F.1	CIFAR-100 train/test	DDPM Easy-to-Hard; shuffled order	Top-1	3	ResNet-18	Eight pairs; full data; 10 bins	Policy sensitivity; matched steps

Table 8: **Complete experiment roadmap: controls and diagnostics.** This continuation maps auxiliary evidence without repeating scalar results.

Location	Dataset / split	Proxy	Target	Gates	Learner	Seeds / budgets	Result / purpose
Tables 33, 34, 31	CIFAR-100 train/test	Balanced DDPM/CE Hard; Random	Top-1; coverage; CE tails	3	ResNet-18	10/30%; reported pairs	Restore counts; retain concentration
Table 32	CIFAR-100 train/test	Scalar DDPM/CE mixtures	Top-1	3	ResNet-18	Three seeds; 10/50/70%	Operating-point diagnostic
Figure 6	CIFAR-100 train	Diffusion timestep sweep	CE, GradNorm	1	ResNet-18 targets	Fixed timestep grid	CE weak across timesteps
Figure 7	CIFAR-100 train	Generative / discriminative regions	Visual regions	Diag.	Source scores	Fixed hardness regions	Distinct sample regions
Table 35	CIFAR-100 train	Target rankings	CE stability	1	ResNet-18/50, ViT-S/16, WRN-50-2	Matched samples	Task-aligned target stability
Table 36	ImageNet-1K val	ResNet-18 CE	ResNet-50 CE / triage	1–3	Pretrained ResNet-18/50	50k; top-10% triage	Positive control, not retraining
Table 37	CIFAR-100 train	Shuffled DDIM	CE, margin, GradNorm	1–2	ResNet-18	Five shuffle seeds	Negative control
Table 38	CIFAR-100 train	Recorded diffusion score	20-superclass CE	1	ResNet-18	Relabeled target	Label-granularity check
Tables 39, 40; Figure 8	CIFAR-100 train	DDPM, SD-v1.5; DINOv2 comparison	Forgetting, first-learning, correctness	1–2	ResNet-18	Separate run; epoch-wise	Dynamics do not unify targets
Table 22	CIFAR-100 train	DDPM score norm	Source CE	1 aux.	Pretrained ResNet-18	HSIC 5k; 200 permutations; seed 0	Nonlinear diagnostic
Table 10	Reported settings	Generative scores, DINOv2	CE, margin, first-learning	2–3	Reported learners	Four candidate R^2 bands	Final claims unchanged

A Additional COHE protocol details

COHE is meant to be applied as an audit rather than as a single aggregate score. In our experiments, each gate uses the same sample identities whenever possible: a vector of task-agnostic proxy values $D_{\text{gen}}(x)$ is paired with discriminative targets $D_{\text{disc}}(x, y; f)$, and all statistics are computed at the sample level before any subset-selection policy is evaluated.

Dependence metrics. Pearson and Spearman correlations measure linear and rank association. The auxiliary HSIC diagnostic pairs the DDPM score-norm array with source-classifier CE on CIFAR-100 train: HSIC uses a seed-0 random 5,000-sample subset and 200 permutations, whereas its accompanying Pearson/Spearman statistics use all 50,000 matched samples (Table 22). A non-significant test does not establish exact independence.

Predictive validity. Held-out R^2 is more operational than a correlation coefficient because it asks whether D_{disc} can be predicted on unseen samples. We report linear models, nonlinear models, and multi-proxy predictors. Negative held-out R^2 means the fitted predictor is worse than predicting the held-out mean.

Operational transfer. Subset-selection experiments train a classifier on selected training subsets and evaluate test accuracy. Budget denotes the retained fraction of the training set. Paired intervals compare each method against the Random subset with the same budget and seed.

A.1 Worked example: threshold calibration in the audit specification

Calibration procedure for a new audit.

```
spec <- record(setting, direction, seeds, baselines, candidates, margin)
null, variability <- matched estimates(spec)
for threshold in spec.candidates:
  if shuffled behavior passes G2 or G3: reject threshold
  else require signal > null + margin and > variability + margin
freeze(spec); compute gates with retained thresholds
G3 pass <- endpoint-specific gain vs baseline; protocol-stable
return tier(gates), sensitivity, scope, pilot status
```

Auditor choices remain explicit; this procedure does not eliminate discretion.

Table 9 relates the retained evidence to this prospective procedure. The historical records support the displayed null checks and descriptive bands, but do not recover a predeclared numerical G1 threshold or G2 safety margin for every setting; no missing threshold is inferred here.

Table 9: **Worked audit-specification example.** Threshold choices are documented and interpreted relative to null and baseline behavior.

Specification field	Retained evidence / procedure	Consequence
Setting declaration	Proxy, target, protocol, direction, splits, and baselines are documented; historical numerical threshold predeclaration is incomplete.	A new audit should freeze its specification before tier assignment.
Gate 2 null	Shuffled-proxy CE/margin R^2 is near zero and split variability is small (Table 37).	CE/margin generative-proxy R^2 values in this band fail predictive validity.
Subset-retraining G3	Paired deltas are evaluated against seed-matched Random with CIs.	This endpoint passes only with a stable positive paired CI.
DINOv2 boundary	First-learning has $\rho = 0.4363$, $R^2 = 0.2126$; direct Hard at 10/30/50% and balanced 10% have 20 negative paired differences.	Target-specific relation; no positive transfer for these policies.
Task-aligned anchor	Cross-architecture rank agreement is strong ($\rho = 0.856\text{--}0.914$).	Positive Gate 1 evidence is preserved, while G2/G3 require matching sample-level audits.

The reported claim readings are unchanged over descriptive R^2 candidate bands $\{0.001, 0.0025, 0.005, 0.010\}$ (Table 10). This is a descriptive sensitivity comparison, not a reconstruction of historical gate decisions or proof of threshold predeclaration.

Table 10: **Threshold-sensitivity check.** Nearby Gate 2 R^2 bands do not change the CE/margin generative-proxy conclusion.

Target and proxy group	Reported R^2	Sensitivity over bands	Tier conclusion
CE loss, generative multi-proxy	0.0002	Below all tested bands.	No reliable transfer; unchanged.
Margin hardness, generative multi-proxy	0.0034	Above 0.001 and 0.0025 only; below 0.005 and 0.010.	No reliable transfer; unchanged because a descriptive band alone is not a gate pass, G1 is weak, and G3 is unsupported.
DINOv2 isolation, first-learning	0.2126	Above all tested bands.	Target-specific relation; unchanged because matched Gate 3 fails.

A.2 Relationship to prior evaluation frameworks

Coleman et al. [2020] assess whether a proxy-based selector matches full-data training performance. COHE additionally asks whether the evidence supports a scalar-surrogate claim for the stated target. The two evaluations are complementary.

Table 11: **Prior-framework comparison.** COHE complements existing validation and proxy-selection frameworks by separating the evidence level from the downstream claim.

Prior framework	What it decides	What it leaves open
Prentice/Freedman surrogate endpoint validation	Population-level surrogate endpoint validity.	Sample-level ML selection actionability and proxy-target-protocol claim tiering.
Coleman et al. (2020) / proxy selection	Whether a proxy-based selector can match full-data training efficiency or downstream accuracy under a selection setup.	Whether dependence, held-out prediction, and downstream transfer are interchangeable evidence levels.
Correlation-only proxy analyses	Scalar rank or linear alignment.	Held-out predictive validity and downstream actionability.
COHE	Which claim tier each evidence pattern licenses for the stated audited setting.	Proxy-level verdicts beyond the declared target, learner, dataset, and protocol scope.

B COHE audit checklist and reporting template

COHE is designed as a reusable audit template across proxy families and discriminative targets; this paper instantiates the audit in standard vision classification. Table 12 gives the minimum reporting template for applying the audit to a new task-agnostic hardness proxy. The checklist is deliberately evidence-oriented: it asks what must be reported before a proxy is described as a discriminative difficulty surrogate.

To apply COHE, provide sample-indexed proxy and target values, held-out split IDs, and optional operational results and budgets. The artifact provides code for the gate analyses, illustrative examples, seven CIFAR-100 proxy scalar arrays with sample-index mappings, and aggregate and seed-level summaries. Table 42 specifies coverage.

Table 12: **COHE audit checklist.** Minimum fields for a reusable proxy audit.

Audit item	Minimal fields to report	Reported evidence
Proxy definition	Model/checkpoint; preprocessing; resolution; score direction.	Sample-indexed D_{gen} ; direction stated.
Target definition	Target type; classifier; label set; checkpoint/epoch; direction.	Sample-indexed D_{disc} for one task notion.
Split protocol	Sample IDs; split seeds; budgets; matched samples.	Audit specification for scoring, prediction, and action.
Gate 1: Dependence	Pearson r ; Spearman ρ ; density/median; optional nonlinear check.	Alignment report.
Gate 2: Prediction	Held-out R^2 ; model class; CI/split variability; proxy set.	Predictive-validity report.
Gate 3: Transfer	Matched baselines; endpoint uncertainty; budgets; diagnostics.	Operational-transfer report.
Claim tier	Tier from Table 13; scope stated.	Bounded claim card.

The six claim tiers use the canonical names in Section 4. Apply Validated surrogate when all gates pass for the stated setting, then Operational-only effect for positive G3 when G1 and G2 do not both pass. When only relational evidence is supported, use Target-specific relation for signal confined to an auxiliary target (including a G1-only GradNorm association), otherwise Predictive validity only for positive G1/G2. Without matched positive G1/G2, use No reliable transfer for measured weak G2 or an evaluated G3 without reliable benefit. Use Weak evidence for remaining incomplete or inconclusive evidence, including a G1-only alignment anchor. An unmeasured gate is not a measured failure, and failed or unavailable G3 does not erase positive G1/G2 evidence. These categories are bounded reporting claims, not universal thresholds.

Table 13: **COHE claim tiers.** Evidence patterns and permitted wording.

Claim tier	Evidence pattern	Permitted wording
Validated surrogate	G1, G2, G3 positive for the declared setting.	Reliable surrogate within this scope.
Predictive validity only	G1/G2 positive; G3 fails or is unavailable; no auxiliary-target restriction.	Predicts this target; downstream benefit unsupported.
Target-specific relation	Relational signal confined to an auxiliary target, dynamics definition, or learner.	Relation to that target; no unified difficulty or unmatched action claim.
Operational-only effect	G3 positive; G1 and G2 do not both pass.	Policy benefit here; scalar-surrogate validity unsupported.
Weak evidence	Incomplete or inconclusive evidence, including G1-only alignment.	Report measured evidence; stronger claims await testing.
No reliable transfer	No matched positive G1/G2; measured G2 is weak or evaluated G3 lacks reliable gain.	No reliable transfer supported here; scalar-surrogate validity unsupported.

Table 14: **Evidence-scope map.** Each evidence block supports a different claim.

Evidence block	Scope	Supports	Boundary
Dependence audit	CIFAR-10/100; ImageNet-10; ImageNet-1K validation.	Sample-wise/rank alignment.	No downstream-action claim.
Predictive-validity audit	Held-out split prediction.	Whether D_{gen} predicts unseen D_{disc} .	No selection-benefit claim.
Operational-transfer audit	CIFAR-100 selection/ordering; ImageNet-10; ImageNet-1K VAE-hard 30%.	Actionability for stated budgets.	Not proof of scalar-surrogate validity.
Artifact/checklist	Gate-analysis and recovered experiment code; seven CIFAR-100 proxy arrays; aggregate and seed-level results; retained author-trained checkpoints.	Verification and partial regeneration of the corresponding reported analyses.	No raw datasets or upstream weights; historical source/checkpoint gaps remain.

C OHS-XOR diagnostic procedure

OHS-XOR is retained only as a diagnostic probe of the two-axis hardness plane. It is not treated as a subset-selection contribution, is not tuned as a competitive policy, and should be read only as a way to test whether off-diagonal regions of the hardness plane behave differently from one-axis rankings.

1. **Input:** dataset $\mathcal{D}$ and subset size k .
2. Compute $D_{\text{gen}}(x)$ and $D_{\text{disc}}(x, y; f)$ for every $(x, y) \in \mathcal{D}$.
3. Set thresholds τ_{gen} and τ_{disc} using fixed percentiles; the experiments reported here use the 75th percentile.
4. Define $\mathcal{H}_{\text{gen}} = \{x : D_{\text{gen}}(x) > \tau_{\text{gen}}\}$ and $\mathcal{H}_{\text{disc}} = \{x : D_{\text{disc}}(x, y; f) > \tau_{\text{disc}}\}$.
5. Form the diagnostic pool
$$\mathcal{S}_{\text{XOR}} = (\mathcal{H}_{\text{gen}} \setminus \mathcal{H}_{\text{disc}}) \cup (\mathcal{H}_{\text{disc}} \setminus \mathcal{H}_{\text{gen}}).$$
6. Uniformly sample k examples without replacement from $\mathcal{S}_{\text{XOR}}$. If the pool has fewer than k examples, retain the full pool and fill the remaining slots uniformly without replacement from its complement.

This procedure tests whether off-diagonal coverage of the hardness plane changes downstream behavior. The paired results in the main paper show mixed behavior, so OHS-XOR should be read as an interpretability probe rather than a competitive sampling rule.

D Formal proxy definitions

The following definitions specify the proxies and targets used in the audit. Each setting retains its own source classifier and score definition; statistics from different source-target arrays should

not be merged. Table 15 lists the eight primary generative proxy families. Table 16 lists auxiliary components, stress tests, and discriminative targets. The CIFAR-100 ConvAE is a dataset-specific reconstruction check, the CLIP row records the semantic component used by the DDIM-CLIP proxy, and DINOv2 is a non-generative representation-level stress test rather than part of the eight primary generative families. The eight primary generative proxy families are DDPM score norm, DDIM reconstruction, DDIM-CLIP, VAE reconstruction, SD-VAE latent round-trip, SD-v1.5 latent denoise, Emu3 VQ reconstruction, and ImageGPT token NLL.

Table 15: **Primary generative proxy families.** Model sources, preprocessing, score direction, evaluated splits, and evaluation roles.

Proxy	Source and preprocessing	Score direction and split	Evaluation role
DDPM score norm	google/ddpm-cifar10-32 UNet; 32×32 CIFAR preprocessing; noised at fixed $t = 500$.	Higher predicted-noise norm; stronger score-field response. Split: CIFAR-100 train.	CIFAR-100 dependence audit
DDIM reconstruction	google/ddpm-cifar10-32 with DDIM scheduler; 32×32 fixed-step reconstruction.	Higher pixel MSE; larger reconstruction error. Split: CIFAR-100 train.	CIFAR-100 dependence audit
DDIM-CLIP	DDPM/DDIM with OpenCLIP ViT-B/32; CLIP features of input and DDIM reconstruction.	Higher $1 - \cos$; larger semantic mismatch. Split: CIFAR-100 train.	CIFAR-100 dependence audit
VAE reconstruction	stabilityai/sd-vae-ft-mse; ImageNet resize/crop; encode/decode pixel reconstruction.	Higher pixel MSE; larger autoencoder error. Split: ImageNet-10 train/val; ImageNet-1K train/val.	ImageNet dependence and transfer audits
SD-VAE latent round-trip	stabilityai/sd-vae-ft-mse; CIFAR/ImageNet encode, decode, and re-encode latent round trip.	Higher latent MSE; less stable latent code. Split: CIFAR-100 train, ImageNet-1K val.	CIFAR-100 and ImageNet dependence audits
SD-v1.5 latent denoise	runwayml/stable-diffusion-v1-5 UNet + VAE; fixed-timestep latent noise prediction after VAE encoding.	Higher latent denoising MSE; larger denoising error. Split: CIFAR-100 train, ImageNet-1K val.	CIFAR-100 and ImageNet dependence audits
Emu3 VQ reconstruction	BAAI/Emu3-VisionTokenizer; 128×128 VQ decode(encode(x)) reconstruction.	Higher pixel MSE; larger VQ reconstruction error. Split: CIFAR-100 train, ImageNet-1K val.	CIFAR-100 and ImageNet dependence audits
ImageGPT token NLL	openai/imagegpt-small; 32×32 quantized image-token sequence.	Higher NLL; less autoregressively predictable tokens. Split: ImageNet-1K val.	ImageNet dependence audit

Table 16: **Auxiliary, stress-test, and target components.** These rows are not part of the eight primary generative families.

Component	Source and preprocessing	Score direction and split	Evaluation role
CIFAR-100-trained ConvAE	Lightweight ConvAE; 32×32 CIFAR-100 train reconstruction MSE.	Higher pixel MSE; larger dataset-specific reconstruction error. Split: CIFAR-100 train.	Dataset-specific reconstruction control
CLIP semantic component	OpenCLIP ViT-B/32; CLIP transform with feature-space cosine discrepancy.	Higher $1 - \cos$; larger semantic discrepancy. Split: CIFAR-100 train.	Semantic-discrepancy dependence audit
DINOv2 feature-space isolation	facebook/dinov2-small; CIFAR-100 train images encoded as CLS embeddings and normalized to unit length.	Higher mean cosine distance to $k = 20$ nearest neighbors; more isolated in representation space. Split: CIFAR-100 train.	Representation-isolation stress test
Discriminative targets	ResNet-18/50 [He et al., 2016], ViT-S/16 [Dosovitskiy et al., 2021], WRN-50-2 [Zagoruyko and Komodakis, 2016]; CE, margin, GradNorm, forgetting, first-learning.	Higher target score means harder unless noted. Split: CIFAR-100 train, ImageNet-1K val.	Dependence, prediction, and control analyses

DDPM denoising and score norm. For a noised sample $x_t = \sqrt{\bar{\alpha}_t}x + \sqrt{1 - \bar{\alpha}_t}\epsilon$, the DDPM denoising-error proxy is

$$D_{\text{gen}}^{\text{DDPM}}(x) = \mathbb{E}_{t, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \right].$$

The DDPM score-norm variant used by Gen-Hard and full-data ordering is $\|\epsilon_\theta(x_{500}, 500)\|_2$ from the CIFAR-10-trained `google/ddpm-cifar10-32` UNet applied to CIFAR-100, using one Gaussian-noise realization per image and higher-is-harder direction. Its formal definition differs from denoising-error MSE; both reflect generative objectives rather than task-semantic difficulty. Stability across noise draws was not measured, so these results do not establish noise-averaged ranking reliability.

DDIM reconstruction and DDIM-CLIP. DDIM reconstruction hardness uses a fixed-step estimate $\hat{x}_0$ and computes

$$D_{\text{gen}}^{\text{DDIM}}(x) = \|x - \hat{x}_0\|_2^2.$$

The semantic variant replaces pixel MSE with CLIP-space discrepancy,

$$D_{\text{gen}}^{\text{DDIM-CLIP}}(x) = 1 - \cos(\text{CLIP}(x), \text{CLIP}(\hat{x}_0)).$$

VAE and latent proxies. The VAE reconstruction proxy is

$$D_{\text{gen}}^{\text{VAE}}(x) = \|x - \text{Dec}(\text{Enc}(x))\|_2^2.$$

For the dataset-specific CIFAR-100 check, we train a lightweight convolutional autoencoder on CIFAR-100 train and use per-sample reconstruction MSE on the same training split as the reconstruction-hardness proxy. For SD-VAE latent round-trip checks, we compute

$$D_{\text{gen}}^{\text{SDVAE}}(x) = \text{mean} \left((z(x) - z(\text{Dec}(z(x))))^2 \right),$$

where $z(\cdot)$ is the frozen SD-VAE encoder. For SD-v1.5 latent denoising, x is encoded into the latent space, noised at a fixed timestep, and scored by latent noise-prediction MSE using the frozen SD-v1.5 UNet.

VQ and autoregressive proxies. The Emu3 VQ-family proxy is pixel reconstruction MSE after `decode(encode(x))` from the pretrained Emu3 vision tokenizer with fixed 128×128 preprocessing. The ImageGPT proxy is mean next-token negative log-likelihood over quantized 32×32 image tokens.

DINOv2 representation-isolation proxy. For the representation-level stress test, we extract DINOv2 CLS features $\phi(x)$, normalize them to unit length, and define

$$D_{\text{rep}}^{\text{DINOv2}}(x_i) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} (1 - \phi(x_i)^\top \phi(x_j)),$$

where $\mathcal{N}_k(i)$ are the $k = 20$ nearest neighbors excluding x_i . Labels do not construct the proxy. We audit this representation-space isolation score against first-learning and the declared hard-only selection policies.

Discriminative targets. The main CE target is $-\log p_f(y | x)$. Margin hardness is $\max_{j \neq y} z_j - z_y$ for logits z : larger values mean harder. GradNorm is the input-gradient norm $\|\nabla_x \mathcal{L}(x, y)\|_2$. Forgetting counts correct-to-incorrect transitions. Here first-learning is the first 1-based epoch after which the sample remains correct through the recorded 50-epoch run; never-stably-learned samples are mapped to 51, with larger values harder.

E Additional dependence results

Table 17: **Main dependence-audit numeric summary.** CE and margin associations remain near zero.

Dataset	Generative proxy	CE		Margin		GradNorm	
		r	ρ	r	ρ	r	ρ
CIFAR-100	DDPM score norm	0.0124	0.0225	0.0260	0.0359	-0.0238	-0.0353
CIFAR-100	DDIM-CLIP	0.0171	0.0524	-0.0477	-0.0510	-0.0648	-0.1307
CIFAR-100	SD-v1.5 latent denoise	-0.0011	-0.0071	0.0278	0.0347	0.0191	0.0649
CIFAR-100	Emu3 VQ reconstruction	0.0009	0.0400	-0.0022	0.0187	-0.0066	-0.1581
ImageNet-1K val	VAE reconstruction	-0.0246	-0.0256	-0.0366	-0.0413	-0.2717	-0.3716
ImageNet-1K val	SD-v1.5 latent denoise	0.0099	0.0105	0.0015	0.0009	-0.2262	-0.2321
ImageNet-1K val	Emu3 VQ reconstruction	-0.0099	-0.0101	-0.0137	-0.0241	-0.0582	-0.3159
ImageNet-1K val	ImageGPT token NLL	0.0087	0.0080	-0.0014	-0.0070	-0.2490	-0.2702

Table 18: **CIFAR-100 dataset-specific reconstruction proxy audit.** CE and margin remain weak for a trained ConvAE.

Target	Dependence	Held-out R^2	Reading
CE loss	$r = 0.0150$	Linear: 0.00019 [0.00002, 0.00036]	No CE-surrogate evidence
	$\rho = 0.0381$	Best: 0.00008 [−0.00021, 0.00037]	
Margin hardness	$r = -0.0095$	Linear: 0.00002 [−0.00023, 0.00027]	No margin-surrogate evidence
	$\rho = -0.0088$	Best: -0.00008 [−0.00017, 0.00000]	
GradNorm	$r = -0.1228$	Linear: 0.01490 [0.01202, 0.01777]	Target-specific sensitivity
	$\rho = -0.2852$	Best: 0.01050 [0.00879, 0.01221]	

Note. R^2 intervals are split-based 95% CIs over five held-out seeds; the best nonlinear model was HistGBDT.

Table 19: **DINOv2 representation-level stress test.** First-learning shows the strongest target-specific signal.

Target	r	ρ	Best R^2	Reading
CE loss	-0.1071	-0.2006	0.0108±0.0014	Weak association; no transfer
Margin hardness	0.0692	0.0852	0.0059±0.0010	Weak association; no transfer
GradNorm	0.1104	0.2669	0.0123±0.0032	Target-specific sensitivity
First-learning epoch	0.4501	0.4363	0.2126±0.0094	Target-specific dynamics
Forgetting count	0.0812	0.0703	0.0359±0.0042	Target-specific dynamics

Note. CIFAR-100 train; proxy is label-free DINOv2 feature-space isolation. Best R^2 reports the better five-split mean between linear regression and HistGBDT; $\pm$ values are split-seed standard deviations.

Table 20: **Auxiliary small-scale dependence checks.** CIFAR-10/100 and ImageNet-10 pairs show weak alignment.

Dataset	Proxy–target pair	n	Pearson r	Spearman ρ
CIFAR-10	DDPM vs CE	50,000	-0.062	-0.076
CIFAR-10	DDPM vs GradNorm	50,000	-0.071	-0.016
CIFAR-100	DDPM vs CE	50,000	0.014	0.007
CIFAR-100	DDPM vs GradNorm	50,000	0.0004	0.002
CIFAR-100	DDIM vs CE	50,000	0.012	0.022
CIFAR-100	DDIM vs GradNorm	50,000	-0.010	-0.013
CIFAR-100	DDIM-CLIP vs CE	50,000	0.058	0.059
ImageNet-10	VAE vs CE	3,925	0.041	0.085

Table 21: **ImageNet-1K validation dependence and predictive checks.** CE and margin remain near zero at scale.

Proxy	Target	Pearson r	Spearman ρ	Linear R^2	Best nonlinear R^2
VAE reconstruction	CE	-0.0246	-0.0256	0.00047	0.00047
VAE reconstruction	Margin	-0.0366	-0.0413	0.00156	0.00156
VAE reconstruction	GradNorm	-0.2717	-0.3716	0.0738	0.1332
SD-VAE latent round-trip	CE	-0.0117	-0.0176	0.00014	0.0000
SD-VAE latent round-trip	Margin	-0.0064	-0.0048	0.00004	0.0000
SD-VAE latent round-trip	GradNorm	-0.0502	-0.1733	0.00252	0.0302
SD-v1.5 latent denoise	CE	0.0099	0.0105	0.00010	0.0000
SD-v1.5 latent denoise	Margin	0.0015	0.0009	0.00000	0.0000
SD-v1.5 latent denoise	GradNorm	-0.2262	-0.2321	0.0512	0.0534
Emu3 VQ reconstruction	CE	-0.0099	-0.0101	0.00010	-0.00009
Emu3 VQ reconstruction	Margin	-0.0137	-0.0241	0.00019	0.00026
Emu3 VQ reconstruction	GradNorm	-0.0582	-0.3159	0.00339	0.1031
ImageGPT token NLL	CE	0.0087	0.0080	0.00008	-0.00014
ImageGPT token NLL	Margin	-0.0014	-0.0070	0.00000	0.00060
ImageGPT token NLL	GradNorm	-0.2490	-0.2702	0.06199	0.06381

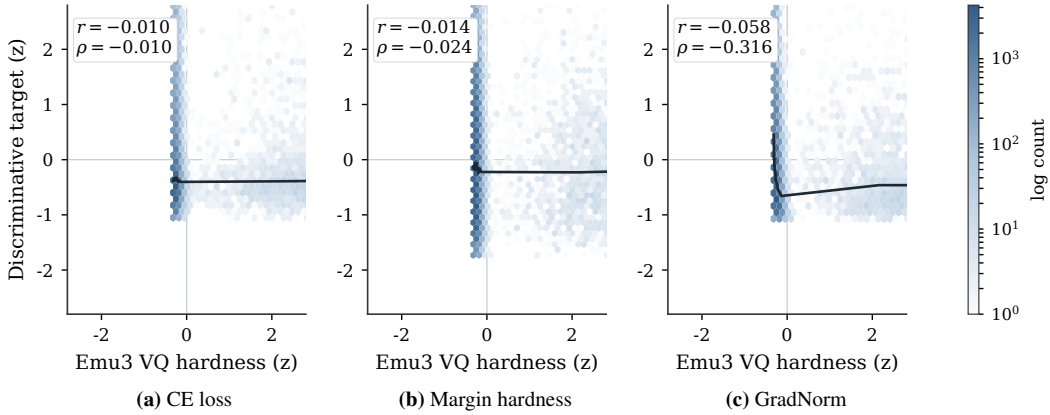


Figure 5: **Emu3 VQ ImageNet-1K hexbin checks.** CE and margin remain diffuse; GradNorm shows target-specific association.

Table 22: **DDPM score-norm versus source ResNet-18 CE on CIFAR-100 train.** Pearson/Spearman use all 50,000 matched samples; HSIC uses a seed-0 5,000-sample subset and 200 permutations.

Metric	Statistic	p -value
Pearson r	-9.47×10^{-5}	0.983
Spearman ρ	-0.00121	0.787
HSIC, RBF kernel	2.94×10^{-5}	0.612

Table 23: **CIFAR-100 predictive-validity split results.** CE and margin prediction remain negligible.

Target D_{disc}	Best single R^2	Multi-proxy R^2	Reading
CE loss	0.0002 ± 0.0004	0.0002 ± 0.0007	Effectively none
Margin hardness	0.0025 ± 0.0011	0.0034 ± 0.0001	Very low
GradNorm	0.0043 ± 0.0014	0.0090 ± 0.0015	Weak target-specific signal
Forgetting count	0.0060 ± 0.0019	0.0083 ± 0.0009	Weak dynamics signal
First-learning epoch	0.0192 ± 0.0026	0.0459 ± 0.0049	Modest, target-specific

Note. Best single denotes the best single-proxy linear fit; multi-proxy uses histogram gradient-boosted trees (HistGBDT). Entries are mean $\pm$ sample standard deviation over split seeds 0–4, not confidence intervals. Figure 3 separately uses 95% CI half-widths $1.96s/\sqrt{5}$. The four inputs are DDIM main, DDIM-CLIP, SD-VAE latent round-trip, and SD-v1.5 latent denoise; 80/20 splits cap fitting at 30,000 samples.

F Operational-transfer details

Table 24: **DINOv2 hard-selection Gate 3.** Five paired seeds per condition on CIFAR-100/ResNet-18. Top-1 is in percent; paired differences and Student- t 95% CIs are in percentage points.

Condition	Hard	Random	Mean Δ	95% CI
Ordinary 10%	10.770	25.640	-14.870	$[-15.616, -14.124]$
Ordinary 30%	26.466	43.720	-17.254	$[-17.986, -16.522]$
Ordinary 50%	40.566	51.684	-11.118	$[-11.536, -10.700]$
Balanced 10%	14.106	25.506	-11.400	$[-12.907, -9.893]$

Table 25: **DINOv2 hard-selection paired differences by seed.** Each entry is Hard minus its seed-matched Random baseline in Top-1 percentage points. The balanced condition balances both arms.

Seed	10%	30%	50%	Balanced 10%
0	-14.35	-18.04	-11.40	-11.46
1	-14.10	-17.01	-11.49	-10.97
2	-15.20	-16.48	-10.65	-13.44
3	-15.28	-17.16	-11.02	-10.29
4	-15.42	-17.58	-11.03	-10.84

The artifact provides the five-seed DINOv2 values and paired summaries. A separate reverse-isolation (DINOv2-Easy) diagnostic used three seeds at 10%: mean Top-1 was 24.25% versus 26.62% for its own Random baseline, with $\Delta = -2.37$ pp (95% CI $[-3.13, -1.61]$). This policy had no corresponding positive G1/G2 evidence and supports No reliable transfer; it is outside the 20 hard-selection pairs above.

F.1 Full-data DDPM blocked ordering

All 50,000 CIFAR-100 training samples are retained exactly once per epoch. The DDPM score-norm proxy from Appendix D defines ten equal-sized bins, traversed Easy-to-Hard with within-bin shuffling. Random uses a global random order; Shuffled-Proxy Order retains the same ten-bin blocked schedule after randomly permuting score-sample correspondence. ResNet-18 is trained from scratch for 200 epochs using SGD (learning rate 0.1, momentum 0.9, weight decay 5×10^{-4}), cosine decay, batch size 128, random crop/flip augmentation, and CIFAR-100 normalization. The three arms use eight paired seeds (0–7), identical optimizer-step counts (78,200), and epoch-200 test Top-1.

Table 26: **Full-data ordering Gate 3.** Paired Student- t 95% intervals and two-sided paired t -tests over eight seeds; units are Top-1 percentage points.

Comparison	Mean Δ	95% CI	p
Easy-to-Hard — Random	-0.77	$[-1.21, -0.33]$	0.0042
Shuffled-Proxy Order — Random	-0.50	$[-0.86, -0.14]$	0.0135
Easy-to-Hard — Shuffled-Proxy Order	-0.27	$[-0.82, 0.28]$	0.2896

The tested fixed blocked-order intervention provides no positive operational transfer. Easy-to-Hard is not statistically distinguished from Shuffled-Proxy Order, so the degradation cannot be uniquely attributed to the true proxy-sample correspondence. This experiment does not test progressive pacing, adaptive curriculum learning, sample weighting, hard-example mining, or other scheduling policies. The artifact provides paired statistics, seed-level results, the fixed ordering protocol, and CPU verification of the ordering. The reported eight pairs use completed runs.

F.2 ImageNet-1K VAE-hard operational selection

The VAE proxy is normalized-pixel reconstruction MSE from `stabilityai/sd-vae-ft-mse`, using posterior mode after `resize/center-crop` to 256 pixels. Scores are computed on 1,281,167 ImageNet-1K training images. For each of 1,000 classes, Hard selects the largest scores and Random samples without replacement; retaining $\lfloor 0.3n_c \rfloor$ per class gives 384,300 training images per arm. Five

seed-matched pairs (0–4) train torchvision ResNet-18 from scratch for 90 epochs with SGD (learning rate 0.1, momentum 0.9, weight decay 10^{-4}), StepLR (step 30, factor 0.1), AMP, and batch size 256. Training uses random resized crop to 224 and horizontal flip; validation uses resize 256/center-crop 224 and ImageNet normalization. All ten runs complete 135,180 optimizer steps.

The primary endpoint uses the same best-validation-Top-1 checkpoint rule for both arms on the full 50,000-image validation split; the final-epoch endpoint is a sensitivity check. Validation is used for checkpoint selection and evaluation. No validation images enter the selection-score source.

Table 27: **ImageNet-1K Gate 3 seed-level Top-1.** Class-balanced VAE-hard selection versus class-balanced Random, 30% budget. Top-1 is in percent; differences are in percentage points.

Seed / endpoint	Proxy-Hard	Random	Paired Δ	Epoch-90 Δ
0	53.812	61.272	-7.460	-7.566
1	53.230	61.482	-8.252	-8.618
2	53.648	61.346	-7.698	-7.774
3	53.698	61.612	-7.914	-7.924
4	53.844	61.346	-7.502	-7.386
Mean, best checkpoint	53.646	61.412	-7.765	-7.854

The mean paired difference is -7.765 pp, with paired Student- t 95% CI $[-8.170, -7.360]$ and two-sided paired t -test $p = 7.47 \times 10^{-7}$. Proxy-Hard is below Random for every seed. Epoch-90 mean difference is -7.854 pp (95% CI $[-8.442, -7.266]$); reported means and differences are rounded from unrounded seed statistics. This larger-scale operational check remains specific to VAE reconstruction, ImageNet-1K, the 30% budget, ResNet-18, class-balanced hard selection, and this training/checkpoint protocol. It does not establish equivalent outcomes for other proxies, architectures, budgets, or interventions.

The artifact provides seed-level results, paired statistics, run and selection checksums, validation of the common checkpoint rule, CPU statistical verification, and the 20 verified author-trained best/final endpoint checkpoints for these ten runs. ImageNet score arrays, images, and upstream pretrained weights are not redistributed.

Table 28: **CIFAR-100 operational-transfer audit.** Paired deltas are reported against seed-matched Random.

Method	Budget 0.1	Budget 0.3	Budget 0.5	Budget 0.7	Reading
Random accuracy	26.71	43.79	51.44	56.31	Five-seed baseline mean
Gen-Hard	+0.93	+0.01	-0.24	-0.43	No consistent transfer
Disc-Hard	-7.92	-9.37	-6.00	-3.26	Hard-only harm
Predictive Entropy	-6.78	-6.35	-3.58	-1.84	Below Random
Coreset-kCenter	-3.58	-2.47	-0.72	-0.47	Mostly below Random
OHS-XOR diagnostic	+0.09	-1.60	-0.85	-0.42	Mixed diagnostic

Note. Non-baseline entries are paired accuracy deltas versus Random over five seeds. Bold negative entries have 95% CIs excluding zero; the Gen-Hard 0.1 positive point is budget-local and not protocol-stable.

Table 29: **Second-learner CIFAR-100 transfer audit.** ResNet-50 retraining gives the same qualitative reading.

Method	Budget			Reading
	0.1	0.5	0.7	
Random acc.	25.25	52.92	59.34	Baseline
Gen-Hard	+0.84 $[-1.06, 2.75]$	+0.13 $[-0.21, 0.48]$	+0.02 $[-0.22, 0.26]$	No consistent transfer
Disc-Hard	-7.83 $[-8.75, -6.91]$	-6.01 $[-6.77, -5.25]$	-3.30 $[-3.90, -2.71]$	Hard-only harm
OHS-XOR diagnostic	+0.16 $[-1.21, 1.52]$	-0.89 $[-1.59, -0.19]$	-0.22 $[-0.64, 0.19]$	Mixed diagnostic

Note. Non-baseline entries are paired accuracy deltas with 95% CIs vs. seed-matched Random over five seeds. This ResNet-50 audit measures 10/50/70%; 30% is not included.

Table 30: **Limited ImageNet-10/Imagenette operational check.** Three-seed deltas are protocol-dependent under a fine-tuning protocol.

Method	Budget		Reading
	0.1	0.3	
Random acc.	96.90	97.73	Baseline
Gen-Hard	-11.48 [−13.66, −9.30]	-0.85 [−1.56, −0.14]	Below Random
Disc-Hard	-8.89 [−10.32, −7.46]	+0.48 [0.42, 0.55]	Budget-dependent
OHS-XOR diagnostic	+0.21 [−0.19, 0.62]	+0.40 [−0.36, 1.16]	Mixed diagnostic

Note. Non-baseline entries are paired accuracy deltas with 95% CIs vs. seed-matched Random over seeds 0–2.

Table 31: **Disc-Hard failure diagnostics.** Hard-only selection concentrates high-loss and joint-hard samples.

Budget	Strategy	Classes	Class entropy	Avg. CE	Top-10% Disc	Joint high-hardness
0.1	Random	100	0.998	4.798	0.100	0.063
0.1	Disc-Hard	94	0.847	5.959	1.000	0.252
0.3	Random	100	1.000	4.793	0.099	0.063
0.3	Disc-Hard	98	0.931	5.492	0.333	0.208

Note. Class entropy is $-\sum_c p_c \log p_c / \log(100)$. Top-10% Disc occupancy follows mechanically from selecting the largest CE values and is a selection-definition check, not independent evidence. CE is measured by the source scoring classifier before subset retraining; exact historical pretrained-weight/head provenance is incomplete.

Table 28 reports the main ranking-policy transfer audit; Table 32 is an auxiliary scalarization sweep only.

Table 32: **α -mix scalarization sweep on CIFAR-100.** Scalarization changes the operating point but not the conclusion.

Budget	$a = 0$	$a = 0.25$	$a = 0.5$	$a = 0.75$	$a = 1$
0.1	18.58±0.09	19.33±0.40	24.71±0.34	27.15±0.11	27.80±0.51
0.5	45.35±0.22	46.19±0.02	49.15±0.15	50.56±0.30	50.89±0.29
0.7	52.90±0.49	53.57±0.28	54.90±0.12	55.82±0.32	56.21±0.10

Note. Select the largest $a z(D_{\text{gen}}) + (1 - a) z(D_{\text{disc}})$, with z-scores over all training samples; $a = 0$ is Disc-Hard and $a = 1$ is Gen-Hard. Entries are test accuracy (%) mean $\pm$ population standard deviation over three seeds. Only 10/50/70% budgets were measured in this historical sweep.

Table 33: **Class-balanced controls.** Class balancing does not yield reliable transfer.

Budget	Method	Method acc.	Bal. Rand.	Δ	95% CI	p
0.1	Balanced Disc-Hard	23.04	26.75	-3.71	[−4.81, −2.61]	0.0007
0.1	Balanced Gen-Hard	27.77	26.75	+1.02	[−0.07, 2.11]	0.0600
0.3	Balanced Disc-Hard	40.06	43.54	-3.48	[−3.69, −3.27]	< 0.0001
0.3	Balanced Gen-Hard	43.84	43.54	+0.30	[−0.18, 0.78]	0.1545

Note. The $p = 0.0600$ point is weak, budget-local evidence because the 95% CI crosses zero and the effect does not persist.

Table 34: **Class-balanced concentration diagnostics.** Class coverage is restored, but tail concentration remains.

Budget	Method	Classes	Class entropy	Mean CE	Top-10% D_{disc}	Joint high-hardness
0.1	Balanced Random	100	1.000	4.792	0.099	0.061
0.1	Balanced Disc-Hard	100	1.000	5.646	0.533	0.209
0.1	Balanced Gen-Hard	100	1.000	4.796	0.100	0.251
0.3	Balanced Random	100	1.000	4.796	0.100	0.062
0.3	Balanced Disc-Hard	100	1.000	5.318	0.279	0.152
0.3	Balanced Gen-Hard	100	1.000	4.793	0.101	0.208

G Diffusion timestep and qualitative checks

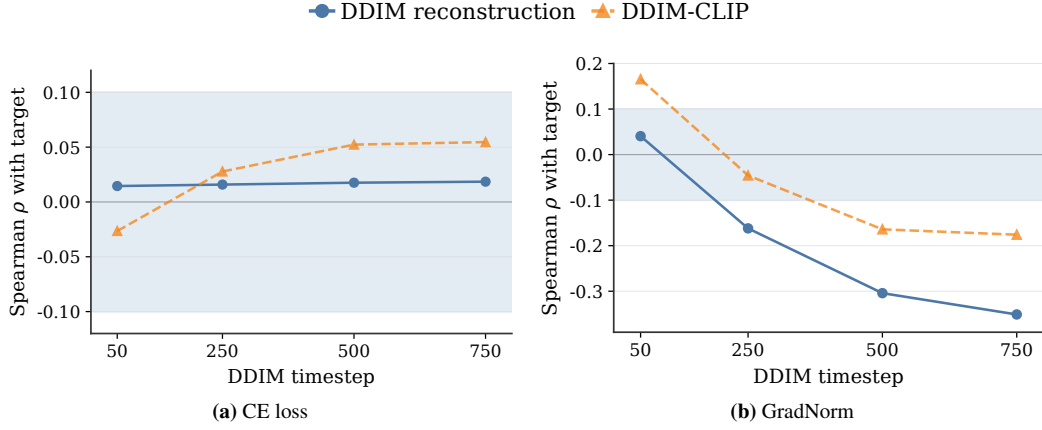


Figure 6: **Diffusion timestep sweep.** CE remains weak across timesteps; GradNorm deviations are proxy-specific.



Figure 7: **Qualitative hardness regions.** Off-diagonal examples illustrate that D_{gen} and D_{disc} can separate different samples.

H Supplementary robustness checks

Table 35: **Cross-architecture consistency of discriminative hardness.** Rankings agree strongly across classifier families.

Architecture	ResNet-18	ViT-S/16	WRN-50-2
ResNet-18	1.000	0.872	0.914
ViT-S/16	0.872	1.000	0.856
WRN-50-2	0.914	0.856	1.000

Table 36: **Positive calibration control.** A task-aligned ImageNet retrieval/triage tuple passes all gates.

Proxy	Target	Gate 1	Gate 2	Gate 3 retrieval	Supported claim
ResNet-18 CE	ResNet-50 CE	$r = 0.752$, $\rho = 0.578$	$R^2 = 0.5637$ [0.5578, 0.5695]; null < 0.0001	Top-10: 62.48% (6.25 $\times$); Top-30: 70.48% (2.35 $\times$)	Validated surrogate for retrieval/triage only

Note. ImageNet-1K validation, $n = 50,000$; target-hard sets are top- q by target CE. Gate 3 evaluates finite-cohort retrieval/triage, not subset retraining. Both recalls exceed the stored shuffled-ranking 97.5th percentiles (10.762% at $q = 0.1$; 30.567% at $q = 0.3$). These are null percentiles, not confidence intervals for recall gains or evidence for retraining benefit.

Table 37: **Shuffled-proxy negative control on CIFAR-100.** Permutation gives near-zero dependence and prediction.

Target	Pearson r	Spearman ρ	Held-out linear R^2
CE loss	+0.0018 $\pm$ 0.0029	+0.0048 $\pm$ 0.0033	-0.0005 $\pm$ 0.0008
Margin hardness	-0.0002 $\pm$ 0.0046	-0.0006 $\pm$ 0.0046	-0.0003 $\pm$ 0.0003
GradNorm	+0.0001 $\pm$ 0.0066	-0.0007 $\pm$ 0.0065	-0.0002 $\pm$ 0.0004

Table 38: **Label-granularity check on CIFAR-100.** Coarser labels do not create strong alignment.

Comparison setting	Pearson r
D_{gen} vs. fine-grained 100-class target	-0.0841
D_{gen} vs. coarse 20-superclass target	-0.0757

Semantic proxy check. To test whether pixel-space reconstruction error drives the low-alignment result, we include DDIM-CLIP as a semantic reconstruction-discrepancy proxy in the main dependence audit. The CLIP-space proxy follows the same low CE/margin surrogate-validity pattern, indicating that the conclusion is not solely a pixel-MSE artifact.

I Training-dynamics details

The single-proxy audit in Table 39 uses DDPM score norm and one seed-0, 50-epoch CIFAR-100/ResNet-18 dynamics run, with three predictive split seeds (0–2). Table 40 and Figure 8 use DDIM main, SD-VAE, and SD-v1.5 against that run’s epoch-wise correctness and forgetting. Table 23 instead fits four proxy inputs over five held-out splits to the final forgetting and mapped first-learning targets; its multi-proxy value is not the single-DDPM result.

Table 39: **CIFAR-100 training-dynamics targets.** The primary proxy has no direct dynamics signal in this run.

Target	Pearson r	Spearman ρ	Best model	Best held-out R^2
Forgetting count	-0.0053	-0.0064	Mean predictor	0.0000
First-learning epoch	-0.0056	-0.0048	Mean predictor	0.0000

Table 40: **Maximum epoch-wise correlations.** Weak relations persist across training stages.

Metric	Proxy	Max $ r $	Max $ \rho $
Cumulative forgetting	DDIM main	0.0687	0.0704
Cumulative forgetting	SD-v1.5 latent diffusion	0.0777	0.0784
Cumulative forgetting	SD-VAE latent	0.0099	0.0132
Current correctness	DDIM main	0.0752	0.0765
Current correctness	SD-v1.5 latent diffusion	0.1088	0.1060
Current correctness	SD-VAE latent	0.0105	0.0307

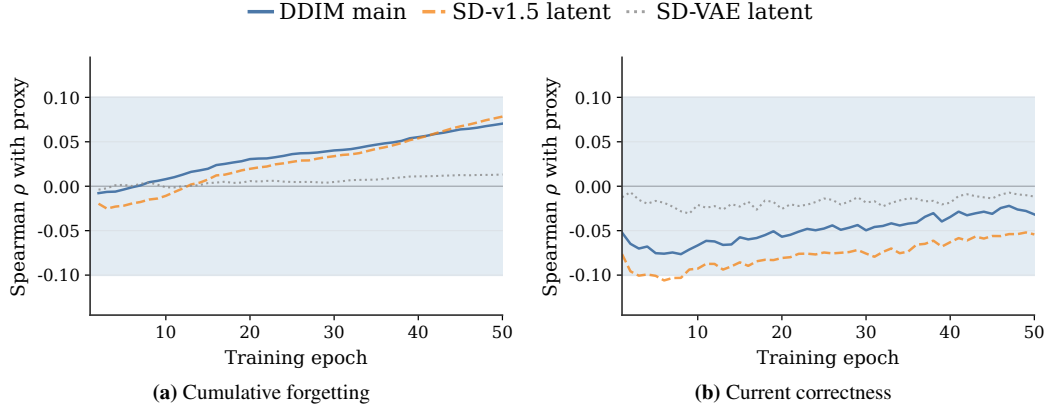


Figure 8: **Epoch-wise training-dynamics correlations.** Spearman correlations remain inside the weak-association band through training.

J Reproducibility

CIFAR-100 subset-selection trains ResNet-18 and repeats the ranking policies with ResNet-50. Predictive fits use five split seeds where reported. ImageNet-1K uses full-validation scoring with pretrained ResNet-50 targets, task-aligned ResNet-18→ResNet-50 retrieval/triage, and the separate five-pair VAE-hard 30% ResNet-18 retraining study (Appendix F.2). ImageNet-10 uses ResNet-18 head fine-tuning at two budgets over three paired seeds. Full-data DDPM ordering uses eight paired seeds (Appendix F.1). Each result retains its own checkpoint and endpoint protocol.

The public COHE artifact supports verification and reuse of the reported analyses. It provides gate-analysis and recovered experiment code, seven CIFAR-100 proxy arrays (DDPM, DDIM, SD-VAE, SD-v1.5, Emu3 VQ, ConvAE, DINOv2) with sample-index mappings, selected targets and split IDs, aggregate and seed-level results, verification utilities, provenance metadata, and 234 verified author-trained checkpoints for retained reported endpoints. Raw datasets, ImageNet-1K score arrays, and third-party pretrained weights are not redistributed. Full regeneration of proxy scores and retraining-based transfer experiments requires the corresponding upstream assets and compute; some historical auxiliary experiments retain partial code or checkpoint coverage.

Table 41: **Artifact scope.** Public resources at release v1.0-neurips2026; full score regeneration and retraining still require upstream assets and compute.

Component	Included?	Reproduction scope
Gate-analysis code	Yes	Computes Gate 1/2/3 summaries from supplied tabular inputs; includes CPU examples and verification utilities.
Recovered experiment code	Yes	Covers the principal released runners; historical gaps are identified in Table 42.
Seven CIFAR-100 scalar arrays	Yes	DDPM, DDIM, SD-VAE, SD-v1.5, Emu3 VQ, ConvAE, and DINOv2, with matched sample-index mappings.
Aggregate and seed-level results	Yes	Authoritative summaries and available per-seed records support reported-result verification.
Author-trained checkpoints	Yes	234 tensor-verified endpoints in nine release archives; missing historical endpoints are listed in Table 42.
Provenance/reproduction documentation	Yes	Records upstream identifiers, checksums, environment evidence, source mappings, and redistribution boundaries.
Raw datasets and third-party weights	No	Obtain from upstream sources under their respective terms; they are not covered by the repository code license.
Complete historical coverage	No	Some runners, scalar inputs, and trained endpoints were not retained; numerical records remain available where stated.

Table 42: **Reproduction coverage by experiment family.** “Ckpt.” counts released author-trained endpoints, not upstream pretrained weights.

Experiment	Scalars/ data	Code	Agg.	Seeds	Ckpt.	Upstream	Status/boundary
Gate 1 dependence	Part.	Yes	Yes	Split rec.	–	Yes	Verification from released inputs; full scoring is partial.
Gate 2 prediction	Part.	Yes	Yes	5 splits*	–	Yes	Verification; some predictive inputs are not redistributed.
CIFAR main Gate 3	DDPM	Yes	Yes	5/cond.	120/120	Yes	Partial; raw CIFAR and upstream scoring model required.
Balanced controls	DDPM	Yes	Yes	5/cond.	30/30	Yes	Partial; upstream data/model required.
Alpha-mix control	Part.	Yes	Yes	3/cond.	45/45	Yes	Partial; upstream inputs required.
ConvAE proxy	Yes	Yes	Yes	Part.	1/1	CIFAR	Verification/scoring code; one retained model.
DINOv2 Gate 3	DINOv2	Yes	Yes	5/cond.	0	Yes	Partial; selection endpoints were not retained.
DDPM ordering	DDPM	Yes	Yes	8 pairs	18/24	Yes	Partial; Random and Easy-to-Hard seeds 0–2 are missing.
ImageNet-10	No	Yes	Yes	3 pairs	0	Yes	Partial; fine-tuning endpoints were not retained.
ImageNet-1K VAE	No	Yes	Yes	5 pairs	20/20	Yes	Partial; score arrays/data/third-party weights required.
Gate 3							
Task-aligned positive control	No	Yes	Yes	Split/null	–	Yes	Partial; upstream ImageNet and target inputs required.
Shuffled negative control	Part.	No*	Yes	Summary	–	Yes	Exact sample-wise historical runner not recovered.
Cross-architecture control	Agg.	No*	Yes	No	0	Yes	Exact historical runner and target endpoints unavailable.
Timestep sweep	Part.	Part.	Yes	Part.	–	Yes	Verification only; exact historical provenance incomplete.
Training dynamics	Part.	Yes	Yes	Part.	0	CIFAR	Verification only; diagnostic target endpoints unavailable.

Note. Part.=partial; Agg.=aggregate only; rec.=records. *Five-split records are available where reported; the full predictive suite is incomplete. “No*” marks an exact historical runner not recovered, not absence of numerical evidence. Seven CIFAR arrays have 50,000 entries and sample mappings. Raw datasets and third-party pretrained weights are never redistributed.

J.1 Artifact scope

Table 42 distinguishes verification from full regeneration and records source and checkpoint gaps. DINOv2 hard-selection comparisons use five paired seeds per condition; the separate three-seed reverse-isolation diagnostic is reported independently. The public artifact is available at <https://github.com/DayenaJeong/cohe-artifact>; the fixed release is v1.0-neurips2026. Tables 41–42 summarize its exact scope.

K Compute and asset provenance notes

Compute resources. Tabular dependence, prediction, and statistical verification use CPU analyses after scores are available. GPU scoring and retraining dominate regeneration costs. ImageNet-1K includes ten completed 90-epoch ResNet-18 runs: each uses one RTX 3090 (24 GiB), scheduled on a workstation with four GPUs, 36 logical CPU cores (Intel i9-10980XE), 188 GiB RAM, and local NVMe storage. Training and validation time is approximately 7.9–17.3 hours per run and 150.2 GPU-hours across these ten runs; this excludes proxy scoring, data preparation, checkpoint I/O, and preliminary/failed work. The training environment uses Python 3.10.19, PyTorch 2.5.1+cu124, torchvision 0.20.1+cu124, and CUDA 12.4. Complete hardware/time totals for scoring and earlier CIFAR/ImageNet-10 studies are unavailable. These figures cover the ten completed ImageNet runs; a full-project compute total is unavailable.

Asset provenance and license status. Table 43 records all external asset families used, including DINOv2 and auxiliary target architectures. License names below come from checkpoint metadata or upstream license files; the repository code license does not establish dataset or third-party-weight redistribution rights. Exact historical OpenCLIP and auxiliary target checkpoint provenance, and the ImageGPT metadata discrepancy, remain documented limitations. No upstream images or weights are redistributed.

Table 43: **Asset provenance.** The paper does not redistribute upstream datasets or model weights.

Asset	Role	Redistributed?	Terms
CIFAR-10/100	Scoring, prediction, transfer	No images	Maintainer dataset page; no explicit license recovered
ImageNet-1K	Validation scoring; train selection and retraining	No images	ImageNet access agreement; research/education conditions
Imagenette / ImageNet-10	Train selection; validation; head fine-tuning	No images	Imagenette source; Apache-2.0 code license; image access terms remain applicable
Google DDPM / DDIM	google/ddpm-cifar10-32; score norm/reconstruction	No weights	Model card: Apache-2.0
SD-VAE	stabilityai/sd-vae-ft-mse; pixel/latent MSE	No weights	Model card: MIT
SD-v1.5	runwayml/stable-diffusion-v1-5; latent denoising	No weights	CreativeML OpenRAIL-M in checkpoint metadata
OpenCLIP	ViT-B/32 semantic discrepancy	No weights	MIT code; exact pretrained tag/weight terms unresolved
Emu3 VisionTokenizer	BAAI/Emu3-VisionTokenizer; VQ reconstruction	No weights	Apache-2.0 in model metadata
ImageGPT	openai/imagegpt-small; token NLL	No weights	Modified MIT upstream; HF Apache-2.0 metadata discrepancy requires confirmation
DINOv2-small	facebook/dinov2-small; isolation	No weights	Apache-2.0 code and model weights
ResNet-18/50	Pretrained targets; author-trained selection/triage learners	Upstream weights: No; retained author endpoints: Partial	torchvision BSD-3-Clause code; timm Apache-2.0 code; release manifest separates author checkpoints
ViT-S/16; WRN-50-2	Cross-architecture target stability	No checkpoints	Exact checkpoint source/training provenance and weight terms unresolved
ConvAE	CIFAR-100-trained reconstruction control	One author-trained checkpoint	Public distribution approved; CIFAR source conditions remain separate

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